

Adaora Mbanefo

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EDUCATION

University of Illinois Urbana-Champaign

B.S. in Statistics and Linguistics; Minor in Computer Science

GPA: 3.8/4.0

Selected coursework: Computer Systems, Data Structures & Algorithms, Statistical Modeling, Statistics & Probability, Linear Algebra, Syntax, Semantics & Pragmatics

In Progress: Applied Machine Learning, Statistical Learning, Computational Linguistics, Algorithms & Models of Computation

RESEARCH EXPERIENCE

ChatterLab, University of Chicago

Junior Researcher (Prof. Marisa Casillas)

Chicago, IL

June 2026 – Present

- Detect children's entry into multiparty conversation in naturalistic daylong recordings, using transformer-based measures of linguistic alignment to mark interaction boundaries
- Build processing pipelines in ALIGN 2.0 (Python) and an ELAN preprocessing converter to bring unstructured recordings into an analyzable format
- Apply speaker diarization to match raw audio to transcripts and construct ground-truth annotations for automated classification

Learning and Language Lab, University of Illinois (NCSA SPIN Program)

Research Assistant (Prof. Jessica Montag)

Champaign, IL

August 2025 – May 2026

- Contributed to the development of a deep learning model in PyTorch combining BERT-derived, visual, and linguistic features with LSTMs to predict eye movements during reading
- Expanded and curated an eye-tracking corpus from 225k to over 800k word fixations across skilled readers
- Applied NLP methods to study question-elicited speech in naturalistic child language environments

TEACHING

Siebel School of Computing and Data Science, University of Illinois

Course Assistant, CS 340: Introduction to Computer Systems

Champaign, IL

August 2026 – Present

PROJECTS

Littafin Fasaha

Learning Engine & Interactive Coding Contributor

Remote

May 2026 – Present

- Built a platform delivering interactive computer science instruction in Hausa
- Fine-tune language models to extend instructional support to other underrepresented African languages
- Develop full-stack features in React/Next.js (TypeScript) with a Supabase/SQL backend, integrating model outputs into an interactive exercise interface

INDUSTRY EXPERIENCE

Inhance Supply Chain Solutions

Data Science Intern

Johannesburg, South Africa

June 2025 – August 2025

- Built an automated Python forecasting pipeline (Darts, StatsForecast) integrated with AWS Redshift and Lambda, deployed across 400+ warehouses

Adanian Labs, AI Center of Excellence

Artificial Intelligence Engineering Intern

Nairobi, Kenya

July 2024 – October 2024

- Co-designed Mwalimu AI, a curriculum planning tool for Kenyan educators, and fine-tuned chatbot backend services for the Agricare product

TECHNICAL SKILLS

Languages: Python, C/C++, Java, Swift, Dart

ML and NLP: PyTorch, TensorFlow, scikit-learn, Hugging Face, spaCy, NumPy, Pandas

Tools: Git, React/Next.js, Supabase/SQL, Flutter

Spoken languages: English (fluent), French (fluent), Sesotho (fluent), Mandarin (intermediate)

SERVICE AND LEADERSHIP

National Center for Supercomputing Applications (NCSA)

Research Conference Committee, Industry Networking Lead

Champaign, IL

October 2025 – April 2026

- Coordinated industry participation with university partners, developed panel questions for the 2026 conference theme, and moderated industry panel discussions

HM Pitje Foundation

Project Manager, Advisory Board, Volunteer

Johannesburg, South Africa

July 2022 – Present

- Organized book drives raising \$3,000+ and supporting digital skills training for 500+ youth in Soshanguve

CONFERENCE PRESENTATIONS

- Mbanefo, A. *Detecting Children's Entry Into Multiparty Conversation Through Linguistic Alignment*. International Congress of Infant Studies, Founding Generation Symposium, September 2026.
- Mbanefo, A., et al. *Question-Elicited Speech Across Naturalistic Language Learning Environments*. Undergraduate Research Symposium, University of Illinois, April 2026.